

Q.P. Code : 12826

(2½ Hours)

[Total Marks : 75

- N.B. : (1) All questions are compulsory.
 (2) Figures to the right indicate full marks.
 (3) Draw neat diagrams wherever necessary.
 (4) Symbols have their usual meanings unless otherwise stated.
 (5) Use of logtable and nonprogrammable calculator is allowed.

1. (a) Attempt any one :

10

- (i) Set up steady state Schrodinger's equation for H-atom in spherical polar co-ordinates. Solve it by the method of separation of variables. Explain how magnetic quantum number m_l arises in solving ϕ equation.
 (ii) Prove that the operators L_z and L^2 have the eigenvalues $m_l \hbar$ and $l(l+1)\hbar^2$ respectively when operated on hydrogen atom wave function.

$$\text{Given } \hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}$$

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]$$

(b) Attempt any one :

5

- (i) Set up steady state Schrodinger's equation for one dimensional harmonic oscillator. Reduce it to dimensionless form by suitable change of variables.
 (ii) What are raising and lowering operators? The n^{th} eigenfunction of a one dimensional linear harmonic oscillator is given by -

$$u_n(y) = \frac{1}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar} \right)^{1/4} e^{-y^2/2} H_n(y)$$

Using the lowering operator, determine the value of n and its energy eigenvalue.

2. (a) Attempt any one :

10

- (i) Show that when the electron jumps from higher energy level E_n to a lower energy level E_m the frequency of photon emitted is

$$\nu = \frac{E_n - E_m}{h}$$

- (ii) Explain with neat diagram, LS and JJ coupling, for two electron atoms.
- (b) Attempt any one :
- What are allowed and forbidden transitions?
 - Determine the spectroscopic terms arising from two electrons in 's' and 'd' states respectively, by LS coupling.
3. (a) Attempt any one :
- What is normal Zeeman effect? Discuss the quantum theory of normal Zeeman effect.
 - Explain Paschen Back effect. Give the quantum mechanical theory of the Paschen - Back effect.
- (b) Attempt any one :
- Find Lande's 'g' factor for $^2P_{3/2}$ and $^2P_{1/2}$.
 - A spectral line of 4000 Å is subject to 0.5 Tesla of magnetic field. The normal Zeeman shift is found to be 0.02795 Å. Determine the specific charge of the electron (e/m). Given $v_e = 3 \times 10^8$ m/s.
4. (a) Attempt any one :
- What is Raman effect? Give quantum mechanical explanation for it.
 - Assuming a diatomic molecule to be a rigid rotator, derive an expression for its moment of inertia and its rotational energy. Explain how energy levels get modified if the effect of bond elongation is taken into consideration.
- (b) Attempt any one :
- The force constant of the bond in CO molecule is 187 N/m. Find the frequency of vibration of the CO molecule and the spacing between the vibrational energy states in terms of electron volt.
 Given: $m(C) = 1.99 \times 10^{-26}$ kg
 $m(O) = 2.66 \times 10^{-26}$ kg
 $h = 6.63 \times 10^{-34}$ J-S
 $1 \text{ eV} = 1.6 \times 10^{-19}$ joule.
- Define polarizability of a molecule. What is polarizability ellipsoid? Explain polarizability of a diatomic molecule.

5. (a) Attempt any one :

4

- (i) Determine the value of zero point energy in eV for a quantum oscillator having period 50 microseconds. Given $h = 6.63 \times 10^{-34}$ JS.
- (ii) Discuss the space quantization of orbital angular momentum for d-state electron in hydrogen atom.

(b) Attempt any one :

4

- (i) What are symmetric and antisymmetric wavefunctions?
- (ii) Explain Hund's rule by considering Mn ($Z = 25$) atom.

5. (c) Attempt any one :

4

- (i) What is anomalous Zeeman effect? State the expression for the additional energy acquired when the atom is placed in a weak magnetic field.
- (ii) Draw the vector model to represent the situation that leads to normal Zeeman effect. Why does normal Zeeman effect occur only in atoms with an even number of electrons?

(d) Attempt any one :

3

- (i) Draw energy level diagram for vibrational rotational Raman spectrum.
 - (ii) Determine the natural frequency of oscillation of Hydrogen molecule if its force constant is 480 N/m and mass of Hydrogen atom is 1.67×10^{-27} kg.
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